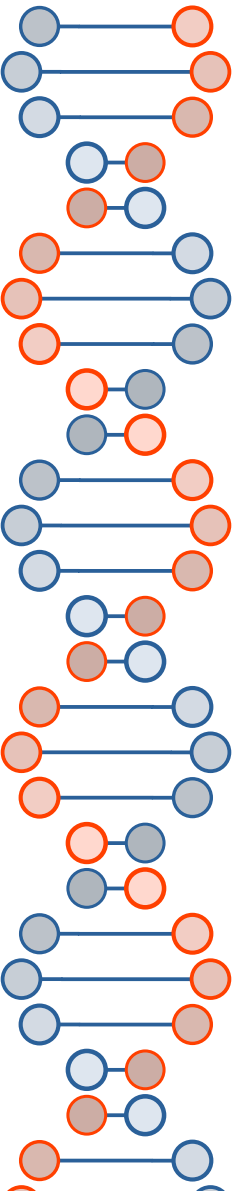


Neural Networks - The Pattern Recognition Engine

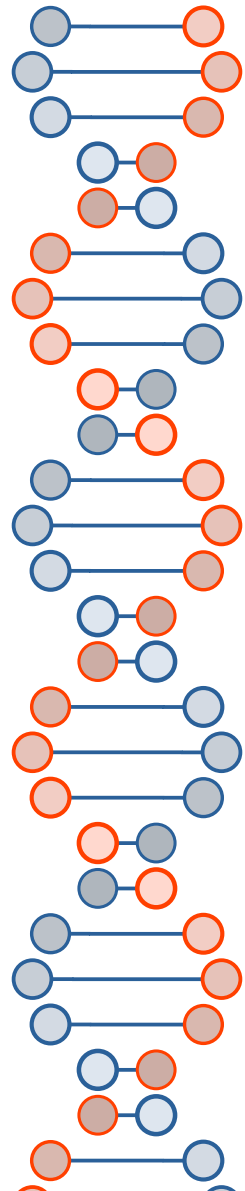
- We now understand how text becomes numerical embeddings that capture meaning and relationships.
- But how do these numbers actually get processed to recognize patterns, understand context, and generate coherent responses?



Neural networks

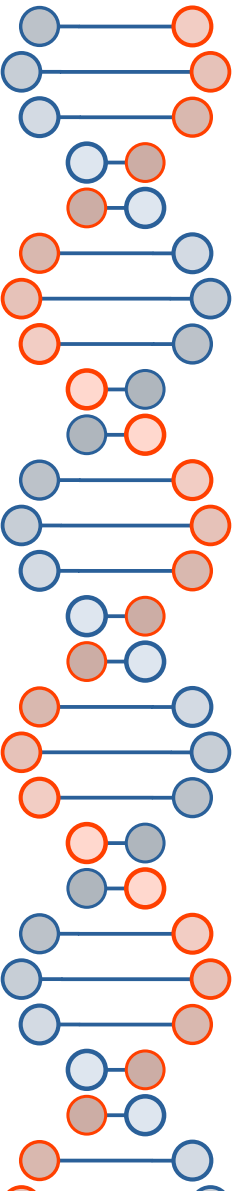
Neural networks: - The sophisticated pattern recognition engines that power modern LLMs.

- Neural networks learn, make predictions, and develop the ability to understand language, all without requiring any mathematical background.
- Think of neural networks as incredibly sophisticated pattern-matching systems that learn from examples, much like how humans develop expertise through experience.



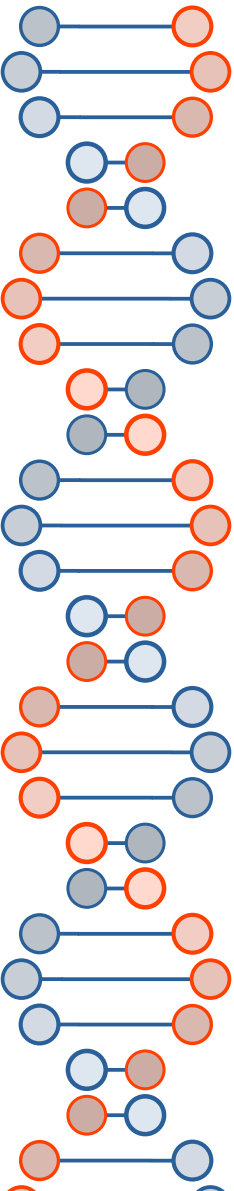
Why Neural Networks for Language?

- Language has a natural hierarchical structure that builds complexity layer by layer:
- **Letters** combine to form **words**
- **Words** combine to form **phrases**
- **Phrases** combine to form **sentences**
- **Sentences** combine to convey **meaning** and **intent**
- Neural networks naturally build understanding in exactly this hierarchical way - simple patterns combine to form complex understanding. This makes them ideal for language tasks where **meaning emerges from the interaction of many smaller components**.



Neural Networks for Language

- Neural networks **take** these **numerical representations** and **process** them **through multiple layers**, with **each layer building a more sophisticated understanding** from the **previous layer's output**.



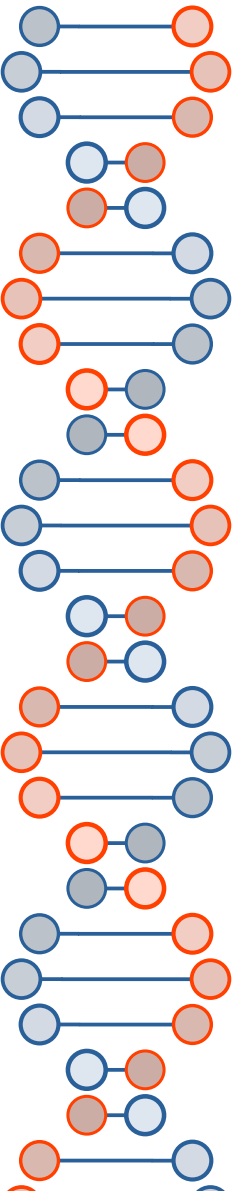
Understanding Neural Networks Through Familiar Analogies

• The Decision Committee Analogy

Imagine you're **deciding whether to approve a business loan**. You might consider several factors:

- **Credit history** (How reliable has this applicant been?)
- **Income stability** (Do they have steady revenue?)
- **Market conditions** (Is their industry growing?)
- **Debt-to-income ratio** (Can they handle additional payments?)

Each factor has a **different importance** in your decision. Some **factors might matter more than others**, and you **combine all this weighted information** to make a final decision: **approve or deny**.

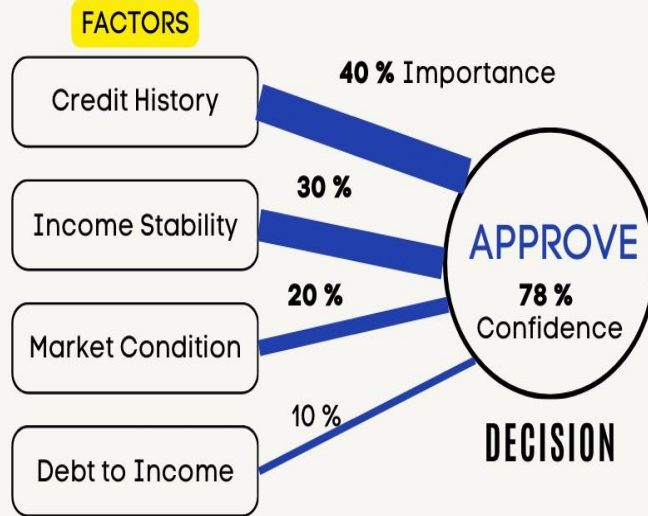


Understanding Neural Networks Through Familiar Analogies

A neural network works Top-k routing (often $k=1$ or 2) keeps FLOP(Floating Point operation)s low while capacity grows.

Similarly, but instead of making loan decisions, it processes numerical inputs (our embeddings) to make predictions about text, such as what word should come next in a sentence.

Business Loan Decision



Neural Network Prediction

